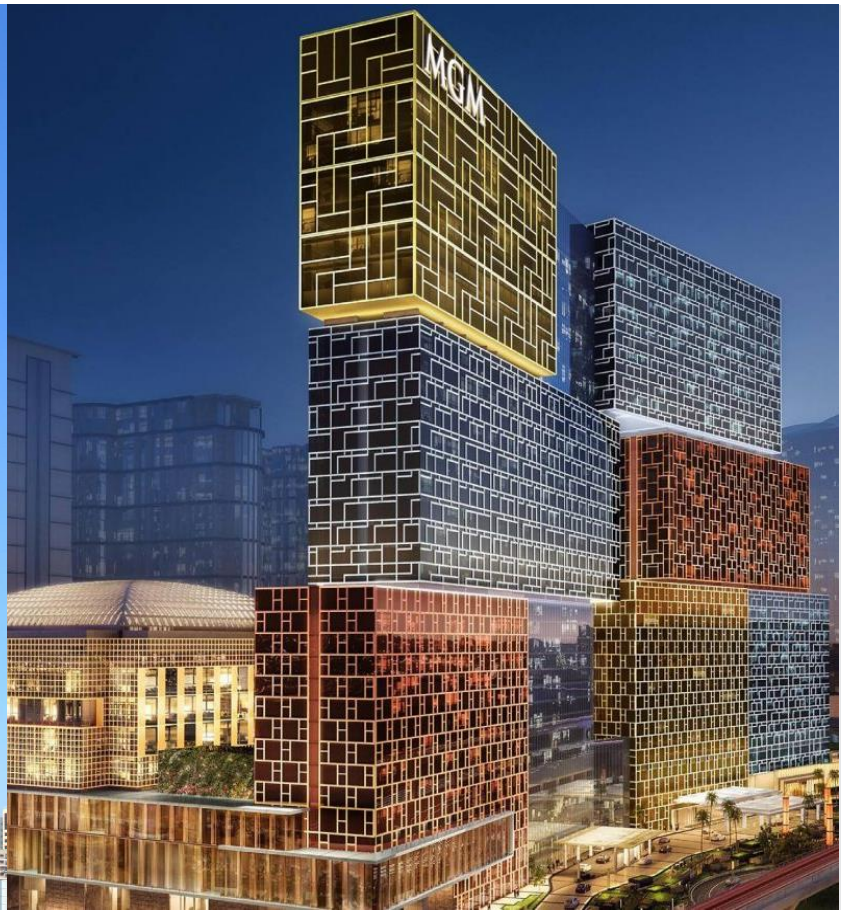
MGM MACAU & COTAI
RESOURCE EFFICIENCY MANAGEMENT REPORT
NOVEMBER 2024

Prepared For:

MGM GRAND PARADISE LIMITED
MGM MACAU, Avenida Dr. Sun Yat Sen Nape, Macau

MGM COTAI, Avenida da Nave Desportiva, Cotai, Macau





NV5 NET ZERO

Unit 5B, 5/F, 133 Wai Yip Street,
Kwun Tong, Hong Kong
Phone: +852 2161 1360

860723-0004415.00
860723-0004367.00

EXECUTIVE SUMMARY

NV5 has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. NV5 appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

NV5 reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as NV5's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Normalized Savings	
MACAU	2019	2025 Target	**-13%**
		Savings as of November 2024	**-20.9%**
COTAI	2019	2025 Target	**-13%**
		Savings as of November 2024	**-6.9%**

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2024.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Savings	
MACAU	2019	2025 Internal Target	**-13%**
		Savings as of November 2024	**+3.1%**
COTAI	2019	2025 Internal Target	**-13%**
		Savings as of November 2024	**+2.0%**

Highlighted Changes in Operation

MACAU

1. MGM MACAU completed the CT-10 fan & motor retrofit pilot project on Jan 2024. Measurement & Verification (M&V) works were conducted to verify the new CT fan and motor performance in Mar 2024.
2. MGM MACAU completed the Natural Gas changeover on 16 Jan 2024.
3. MGM MACAU switched ON three boilers (1 big boiler and 2 small boilers) during 1 Feb – 6 Mar 2024 due to low OAT fluctuation and final testing for NG system.
4. MGM MACAU completed the rooftop façade LED replacement in Feb 2024.
5. MGM MACAU completed the Lion Backdrop lighting replacement in Jun 2024.
6. MGM MACAU has increased casino ACH from 3.0 to 6.0 on 29 May 2024.
7. MGM MACAU has configured the Main Boiler at stand-by mode on 2 – 31 Jul 2024 due to SB01 and SB02 annual maintenance on 2 – 26 Jul 2024.
8. MGM MACAU has turned on the Main Boiler and two small boilers on 23 - 26 Sep 2024 for T&C after replacing the control panel and automatic control system.
9. MGM MACAU has implemented ACH energy saving mode for casino AHU during 3:00am – 9:00am daily operation on 30 Oct 2024.
10. MGM MACAU has completed the new Chiller replacement for WCC-03 in the early Dec 2024.

COTAI

1. MGM COTAI completed CDP Extension Work at B2 Carpark (Phase 11 & 12).
2. MGM COTAI further adjusted the back of house AHUs for night mode low fan speed operation.
3. MGM COTAI continues to replace the back of house office lighting panel with occupancy sensor in different phases.
4. MGM COTAI has resumed the power supply of MDM Kitchen and Restaurant to their original circuits.
5. MGM COTAI has implemented low fan speed night mode (00:00 – 06:00) for Ballroom/Conference Room/Spectacle/Vista AHUs starting from 29 Apr 2024.
6. MGM COTAI has extended the Night Mode low fan speed setting of multiple AHUs for warehouse and security office from 22:00 – 06:00 to 20:00 – 06:00.
7. MGM COTAI has modified and updated the control programs and graphics of the Aji and Chun kitchen exhaust fans in BMS and iBMS.
8. MGM COTAI has increased casino ACH from 4.5 to 5.0 on 28 Jun 2024.
9. MGM COTAI has switched ON NG Boiler due to power outage for Chiller and Heat Pumps on 24 July 2024 12:03. It resumed normal operation on 25 July 2024 17:50.
10. MGM COTAI completed VSD installation for 12 units of exhaust air fan on 7 Aug 2024.
11. MGM COTAI has performed 11kV starter panel maintenance on 9 - 13 Sep 2024, hot water was supplied by NG boiler on 11 Sep 2024.
12. MGM COTAI has implemented ACH energy saving mode for casino AHU during 3:00am – 9:00am daily operation on 1 Nov 2024

Consumption Intensity

The 2024 YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m²)	% Change Compared to 2023
MACAU	**28.5**	**+4.6%**
COTAI	**28.5**	**+2.8%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m³/ m²)	% Change Compared to 2023
MACAU	**0.30**	**+10.7%**
COTAI	**0.21**	**+3.5%**

ISO 50001 Certification Audit

MACAU

The HKQAA 2024 renewal audit was completed on 1 March 2024. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

COTAI

The HKQAA 2024 surveillance audit was completed on 8 November 2024. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

Milestones in Year 2024

The following activities will be completed in year 2024:

Table 5. MGM MACAU Year 2024 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Replacement (Phase 1 : 1 unit)	Completed
2	Cooling Tower Fan Retrofit (Phase 1 : 1 unit)	Completed
3	Cooling Tower Fan Retrofit (Phase 2 : 7 units)	On-going
4	Secondary Chilled Water Pump Retrofit (Phase 1 : 3 units)	On-going
5	AHU EC Fan Replacement (Phase 1 : 6 units)	On-going
6	Neon Light LED Replacement (Phase 3)	Completed
7	Rooftop Façade LED Replacement	Completed
8	Lion Backdrop Lighting Upgrade	Completed
LPG		
9	Natural Gas Changeover Project	Completed

Table 6. MGM COTAI Year 2024 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset…)	On-going
2	PICCV Implementation 2024 (14 units)	On-going
3	AHU/PAU EC Fan Replacement (Phase 1 : 5 units)	On-going
4	Additional VSD for Exhaust Air Fan (Phase 2 : 12 units)	Completed
5	KEF Steps Speed Control Optimization 2024 (Chun, IRD, Hu Dao, Coast, Grill 58, Mansion)	On-going
6	Back of House Offices Lighting Replacement	Completed
7	CRO Booster Pump Replacement (Phase 2 : 15 units)	Completed
Water		
8	Extension of B2 Carpark CDP Water Recycling (Phase 11)	Completed
9	Extension of B2 Carpark CDP Water Recycling (Phase 12)	Completed

TABLE OF CONTENTS

EXECUTIVE SUMMARY 1

ISO 50001 Certification Audit 3
Milestones in Year 2024 3

1.0 PORTFOLIO PERFORMANCE SUMMARY 7

1.1 Electricity Use Intensity (EUI) 7
1.2 Water Use Intensity (WUI) 8

2.0 MGM MACAU PERFORMANCE 9

2.1 Electricity consumption 9
2.2 Gas Consumption 11
2.3 ISO50001 Energy Consumption Record 13
2.4 Water Consumption 14
2.5 MACAU Year 2024 Projects 16

3.0 MGM COTAI PERFORMANCE 17

3.1 Electricity Consumption 17
3.2 Natural Gas Consumption 20
3.3 ISO50001 Energy Consumption Record 22
3.4 Water Consumption 23
3.5 COTAI Year 2024 Projects 25

APPENDIX A 26

Regression Analysis 26
MGM MACAU 26
MGM COTAI 26

APPENDIX B 27

MACAU Gas Consumption Raw Data 27

ACKNOWLEDGMENT

NV5 has been working with MGM Team as the energy and sustainability manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that NV5 has assisted MGM MACAU and COTAI to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, Macao Green Hotel Award and China 3 Star Green Building Operation Label.

NV5 appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1.0 PORTFOLIO PERFORMANCE SUMMARY

1.1 ELECTRICITY USE INTENSITY (EUI)

MACAU's electricity use intensity (EUI) in November 2024 had increased by 3.1% compared to the same month in 2023 and increased by 4.6% in 2024 compared to 2023. Meanwhile, COTAI's EUI in November 2024 decreased by 0.4% compared to the same month in 2023 and increased by 2.8% in 2024 compared to 2023.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2023 Actual EUI (kWh/m²)	Year 2024 Actual EUI (kWh/m²)	% Change	Year 2023 Actual EUI (kWh/m²)	Year 2024 Actual EUI (kWh/m²)	% Change
January	22.4	23.7	+5.4%	22.6	24.2	+6.9%
February	21.2	23.0	+8.3%	21.0	23.0	+9.2%
March	24.6	25.9	+5.4%	24.4	25.9	+6.3%
April	25.0	29.4	+17.7%	25.8	28.9	+12.3%
May	28.5	29.3	+3.0%	29.4	29.5	+0.4%
June	30.3	30.4	+0.1%	31.5	31.5	-0.2%
July	31.9	32.8	+2.8%	32.5	32.8	+1.0%
August	31.6	32.5	+2.7%	32.0	32.7	+2.2%
September	30.3	31.4	+3.7%	30.5	30.5	-0.3%
October	28.7	29.2	+1.7%	29.2	28.8	-1.4%
November	25.2	26.0	+3.1%	25.8	25.7	-0.4%
December						
Average	**27.3**	**28.5**	**+4.6%**	**27.7**	**28.5**	**+2.8%**

1.2 WATER USE INTENSITY (WUI)

MACAU's water use intensity (WUI) in November 2024 decreased by 1.6% compared to the same month in 2023 and increased by 10.7% in 2024 compared to 2023. Meanwhile, COTAI's WUI in November 2024 decreased by 3.0% compared to the same month in 2023 and increased by 3.5% in 2024 compared to 2023.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2023 Actual WUI (m^3/m^2)	Year 2024 Actual WUI (m^3/m^2)	% Change	Year 2023 Actual WUI (m^3/m^2)	Year 2024 Actual WUI (m^3/m^2)	% Change
January	0.19	0.24	+28.7%	0.14	0.17	+19.0%
February	0.21	0.26	+20.4%	0.16	0.18	+9.1%
March	0.21	0.27	+27.8%	0.17	0.17	+5.7%
April	0.24	0.32	+34.0%	0.19	0.21	+12.2%
May	0.28	0.31	+10.4%	0.21	0.20	-5.0%
June	0.29	0.30	+2.3%	0.23	0.20	-11.3%
July	0.32	0.34	+6.2%	0.25	0.26	+2.2%
August	0.34	0.33	-0.6%	0.23	0.25	+8.4%
September	0.30	0.31	+3.5%	0.22	0.22	+0.8%
October	0.30	0.33	+7.5%	0.21	0.23	+9.6%
November	0.30	0.29	-1.6%	0.21	0.20	-3.0%
December						
Average	**0.27**	**0.30**	**+10.7%**	**0.20**	**0.21**	**+3.5%**

2.0 MGM MACAU PERFORMANCE

2.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in November 2024 increased by 3.1% and 1.9% respectively compared to the same month in 2023. To date, MGM MACAU has increased the actual electricity consumption by 4.6% and associated costs increased by 5.0%.

Table 9. MACAU Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2023	2024	% Change Actual	2023	2024	% Change Actual	
January	4,745,400	5,003,520	+5.4%	5,668,020	5,973,270	+5.4%	+173.7%
February	4,488,360	4,860,480	+8.3%	5,365,140	5,812,030	+8.3%	+17.2%
March	5,205,240	5,485,440	+5.4%	6,207,870	6,530,510	+5.2%	+5.6%
April	5,284,800	6,220,200	+17.7%	6,236,780	7,458,080	+19.6%	+50.0%
May	6,019,800	6,200,040	+3.0%	7,106,470	7,425,270	+4.5%	-3.5%
June	6,418,080	6,425,760	+0.1%	8,307,460	8,437,580	+1.6%	-6.1%
July	6,748,440	6,940,080	+2.8%	8,730,530	9,046,120	+3.6%	-1.4%
August	6,692,280	6,870,240	+2.7%	8,647,480	8,959,640	+3.6%	+2.5%
September	6,417,720	6,652,800	+3.7%	8,297,960	8,678,960	+4.6%	+3.7%
October	6,075,360	6,181,560	+1.7%	7,129,400	7,156,090	+0.4%	+8.6%
November	5,339,640	5,506,560	+3.1%	6,263,910	6,381,840	+1.9%	-8.4%
December							
Total	**63,435,120**	**66,346,680**	**+4.6%**	**77,961,020**	**81,859,390**	**+5.0%**	**+5.3%**

***Important note**: Cooling Degree Days (CDDs) and food cover are the main variables taken into consideration for electricity consumption normalization.*

NV5 summarized the key items that could result in an electricity consumption increase below:

- Kitchen electricity consumption has increased by 11.0% in November 2024 compared with the same month in 2023.
- Heat pump plant electricity consumption has increased by 22.9% in November 2024 compared with the same month in 2023.

Table 10 presents a comparison of electricity rates in the years 2023 and 2024. The electricity rate in November 2024 decreased by 1.2% compared to the same month in 2023.

Table 10. MACAU - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2023 – Electricity Rate (MOP/kWh)	Year 2024 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.194	1.194	-0.1%
February	1.195	1.196	+0.04%
March	1.193	1.191	-0.2%
April	1.180	1.199	+1.6%
May	1.181	1.198	+1.4%
June	1.294	1.313	+1.4%
July	1.294	1.303	+0.8%
August	1.292	1.304	+0.9%
September	1.293	1.305	+0.9%
October	1.173	1.158	-1.4%
November	1.173	1.159	-1.2%
December			
Average	**1.224**	**1.229**	**+0.4%**

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in November 2024 decreased by 29.8% compared to 2019 normalized consumption in the same month.

Table 11. MACAU - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2024 Actual Electricity Consumption (kWh)	% Change – Normalized
January	6,303,975	5,003,520	-20.6%
February	6,353,462	4,860,480	-23.5%
March	6,893,236	5,485,440	-20.4%
April	7,544,168	6,220,200	-17.5%
May	8,006,220	6,200,040	-22.6%
June	8,175,745	6,425,760	-21.4%
July	8,913,733	6,940,080	-22.1%
August	9,285,966	6,870,240	-26.0%
September	8,673,105	6,652,800	-23.3%
October	8,238,000	6,181,560	-25.0%
November	7,848,355	5,506,560	-29.8%
December			
Total	**86,235,965**	**66,346,680**	**-23.1%**

2.2 GAS CONSUMPTION

Gas consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants Gas utilization cover over 95% of the overall site Gas consumption, NV5 focuses the analysis on those two areas.

As of 16 January 2024, MGM Macau has transitioned from LPG to Natural Gas for all gas-operated equipment.

According to Table 12, the actual Gas consumption in November 2024 increased by **6.8%** while the cost decreased by **-10.8%** respectively compared to the same month in 2023. To date, MGM MACAU has increased the actual Gas consumption by **14.0%** and associated costs increased by **4.5%.**

Table 12. MACAU – Summary of Actual Monthly Gas Consumption and Cost

Monthly Period	Actual Gas Consumption (GJ)			Actual Gas Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2023 (LPG)	2024 (LPG + NG)	% Change Actual	2023 (LPG)	2024 (LPG + NG)	% Change Actual		
January	5,577	5,760	+3.3%	1,084,579	1,211,135	+11.7%	-37.6%	+50.5%
February	3,555	5,487	+54.3%	768,347	1,052,400	+37.0%	-35.8%	+66.8%
March	3,515	4,619	+31.4%	759,672	885,980	+16.6%	-41.6%	+58.4%
April	3,494	4,088	+17.0%	755,191	784,030	+3.8%	+10.3%	+45.8%
May	3,641	3,780	+3.8%	839,195	725,110	-13.6%	+8.9%	+59.7%
June	3,227	3,398	+5.3%	743,662	651,740	-12.4%	-7.1%	+49.0%
July	3,055	3,456	+13.1%	704,081	662,870	-5.9%	-1.5%	+29.3%
August	3,261	3,511	+7.7%	625,540	673,460	+7.7%	-3.0%	+32.5%
September	3,169	3,505	+10.6%	607,893	672,320	+10.6%	-7.7%	+28.0%
October	3,534	3,714	+5.1%	677,857	712,430	+5.1%	+14.9%	+13.6%
November	3,510	3,749	+6.8%	805,816	719,010	-10.8%	-1.7%	+32.9%
December								
Total	**39,538**	**45,067**	**+14.0%**	**8,371,834**	**8,750,485**	**+4.5%**	**-6.5%**	**+39.9%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration for Gas consumption normalization.*

NV5 summarized the key items which could result in Gas consumption increase below:

- Kitchen Gas consumption has increased by 7.3% due to Food Cover increased by 32.9% in November 2024 compared with the same month in 2023.

Table 13 presents the comparison of Gas rates in the years 2023 and 2024. The Gas rates in November 2024 decreased by 16.5% compared with the same month in 2023.

Table 13. MACAU - Summary of Actual Monthly Gas Rate

Monthly Period	Year 2023 – Gas Rate (MOP/GJ)	Year 2024 – Gas Rate (MOP/GJ)	% Change – Gas Rate
January	194.48	210.27	+8.1%
February	216.11	191.80	-11.2%
March	216.11	191.80	-11.2%
April	216.11	191.80	-11.2%
May	230.46	191.80	-16.8%
June	230.46	191.80	-16.8%
July	230.46	191.80	-16.8%
August	191.83	191.80	-0.01%
September	191.83	191.81	-0.01%
October	191.83	191.80	-0.02%
November	229.58	191.81	-16.5%
December			
Average	**212.66**	**193.48**	**-9.0%**

Table 14 summarizes the monthly normalized Gas consumption. The actual Gas consumption in November 2024 has decreased by 10.7% compared to 2019 normalized consumption in the same month.

Table 14. MACAU - Summary of Normalized Monthly Gas Consumption

Monthly Period	2019 Normalized Gas Consumption (GJ)	2024 Actual Gas Consumption (GJ)	% Change – Normalized
January	5,901	5,760	-2.4%
February	5,135	5,487	+6.9%
March	5,168	4,619	-10.6%
April	4,128	4,088	-1.0%
May	3,872	3,780	-2.4%
June	3,797	3,398	-10.5%
July	4,067	3,456	-15.0%
August	4,155	3,511	-15.5%
September	3,710	3,505	-5.5%
October	4,135	3,714	-10.2%
November	4,200	3,749	-10.7%
December			
Total	**48,267**	**45,067**	**-6.6%**

2.3 ISO50001 ENERGY CONSUMPTION RECORD

The MACAU ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to November 2024 is **20.9%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG/NG combined) compared to the 2019 baseline.

Table 15. MACAU ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Normalized Savings	
MACAU	2019	2025 Target	**-13%**
		Savings as of November 2024	**-20.9%**

Table 16. MACAU ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2024 Actual Energy Consumption (GJ)	% Change – Normalized
January	28,595	23,773	-16.9%
February	28,008	22,985	-17.9%
March	29,984	24,367	-18.7%
April	31,287	26,480	-15.4%
May	32,694	26,101	-20.2%
June	33,230	26,531	-20.2%
July	36,156	28,440	-21.3%
August	37,584	28,244	-24.9%
September	34,933	27,455	-21.4%
October	33,792	25,968	-23.2%
November	32,454	23,572	-27.4%
December			
Total	**358,717**	**283,915**	**-20.9%**

Important note: *Factors used for unit conversion: Electricity (0.0036 GJ/kWh), LPG (0.0453 GJ/kg), and NG (0.0347 GJ/m³)*

2.4 WATER CONSUMPTION

Based on the existing water metering system, the overall site water consumption can be broken down into five (5) categories: Kitchen and Restaurants; Hotel Tower; Fire Services; Chiller Plant & miscellaneous.

According to Table 17, the actual water consumption and cost in November 2024 has decreased by 1.6% and 1.6% respectively compared to the same month in 2023 respectively. To date, MGM MACAU has increased both the actual water consumption and associated cost by 10.7% and 10.6% respectively.

Table 17. MACAU - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2023	2024	% Change Actual	2023	2024	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	39,305	50,588	+28.7%	305,020	392,330	+28.6%	+173.7%	+50.5%
February	45,112	54,323	+20.4%	350,080	421,370	+20.4%	+17.2%	+66.8%
March	44,725	57,148	+27.8%	347,050	443,270	+27.7%	+5.6%	+58.4%
April	49,960	66,952	+34.0%	387,650	519,260	+34.0%	+50.0%	+45.8%
May	58,878	65,014	+10.4%	456,800	504,150	+10.4%	-3.5%	+59.7%
June	61,331	62,753	+2.3%	475,780	486,660	+2.3%	-6.1%	+49.0%
July	68,165	72,403	+6.2%	528,630	561,220	+6.2%	-1.4%	+29.3%
August	71,191	70,758	-0.6%	552,160	548,670	-0.6%	+2.5%	+32.5%
September	64,118	66,358	+3.5%	497,260	514,480	+3.5%	+3.7%	+28.0%
October	64,415	69,273	+7.5%	499,470	536,990	+7.5%	+8.6%	+13.6%
November	63,109	62,127	-1.6%	489,490	481,560	-1.6%	-8.4%	+32.9%
December								
Total	**630,309**	**697,697**	**+10.7%**	**4,889,390**	**5,409,960**	**+10.6%**	**+5.3%**	**+39.9%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration for water consumption normalization.*

NV5 summarized the key items that could result in a water consumption decrease below:

- Hotel water consumption has decreased by 35.2% compared with the same month in 2023.

Table 18 presents the comparison of water rates in the years 2023 and 2024. The water rate in November 2024 has decreased by 0.07% compared to the same month in 2023. The average utility rate of water in 2024 has decreased by 0.04% compared to 2023.

Table 18. MACAU - Summary of Actual Monthly Water Rates

Monthly Period	Year 2023 – Water Rate (MOP/m³)	Year 2024 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.760	7.755	-0.06%
February	7.760	7.757	-0.04%
March	7.760	7.757	-0.04%
April	7.759	7.756	-0.05%
May	7.758	7.754	-0.05%
June	7.758	7.755	-0.03%
July	7.755	7.751	-0.05%
August	7.756	7.754	-0.02%
September	7.755	7.753	-0.03%
October	7.754	7.752	-0.03%
November	7.756	7.751	-0.07%
December			
Average	**7.757**	**7.754**	**-0.04%**

Table 19 summarized the monthly normalized water consumption. The actual water consumption in November 2024 has decreased by 10.9% compared to 2019 normalized consumption in the same month.

Table 19. MACAU - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2024 Actual Water Consumption (m³)	% Change – Normalized
January	57,198	50,588	-11.6%
February	57,605	54,323	-5.7%
March	61,714	57,148	-7.4%
April	66,102	66,952	+1.3%
May	70,123	65,014	-7.3%
June	71,011	62,753	-11.6%
July	76,983	72,403	-5.9%
August	80,240	70,758	-11.8%
September	75,144	66,358	-11.7%
October	71,822	69,273	-3.5%
November	69,716	62,127	-10.9%
December			
Total	**757,658**	**697,697**	**-7.9%**

2.5 MACAU YEAR 2024 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2024 and reduce the site energy and water consumption which supports MGM energy and water savings target.

Table 20. MACAU Year 2024 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Replacement (Phase 1 : 1 unit)	Completed
2	Cooling Tower Fan Retrofit (Phase 1 : 1 unit)	Completed
3	Cooling Tower Fan Retrofit (Phase 2 : 7 units)	On-going
4	Secondary Chilled Water Pump Retrofit (Phase 1 : 3 units)	On-going
5	AHU EC Fan Replacement (Phase 1 : 6 units)	On-going
6	Neon Light LED Replacement (Phase 3)	Completed
7	Rooftop Façade LED Replacement	Completed
8	Lion Backdrop Lighting Upgrade	Completed
LPG		
9	Natural Gas Changeover Project	Completed

3.0 MGM COTAI PERFORMANCE

3.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in November 2024 decreased by 0.4% and 2.0% respectively compared to the same month in 2023. To date, MGM COTAI has increased the actual electricity consumption by 2.8% and associated costs increased by 3.3%.

Table 21. COTAI Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2023	2024	% Change Actual	2023	2024	% Change Actual	
January	8,471,280	9,054,720	+6.9%	10,516,100	11,239,000	+6.9%	+173.7%
February	7,869,600	8,597,400	+9.2%	9,794,150	10,692,600	+9.2%	+17.2%
March	9,128,520	9,700,320	+6.3%	11,334,370	12,045,940	+6.3%	+5.6%
April	9,638,280	10,820,760	+12.3%	11,855,350	13,530,960	+14.1%	+50.0%
May	10,993,680	11,032,920	+0.4%	13,528,490	13,782,190	+1.9%	-3.5%
June	11,802,600	11,777,640	-0.2%	14,491,170	14,717,760	+1.6%	-6.1%
July	12,146,160	12,270,000	+1.0%	14,926,180	15,204,900	+1.9%	-1.4%
August	11,961,480	12,230,400	+2.2%	14,684,180	15,163,300	+3.3%	+2.5%
September	11,428,920	11,393,400	-0.3%	14,051,860	14,131,730	+0.6%	+3.7%
October	10,939,680	10,790,160	-1.4%	13,353,730	12,953,220	-3.0%	+8.6%
November	9,667,440	9,630,120	-0.4%	11,823,780	11,587,110	-2.0%	-8.4%
December							
Total	**114,047,640**	**117,297,840**	**+2.8%**	**140,359,360**	**145,048,710**	**+3.3%**	**+5.3%**

***Important note**: Cooling Degree Days (CDDs), Food Cover, Room Night and E-Bus Mileage are the main variables taken into consideration on electricity consumption normalization.*

NV5 summarized the key items which could result in an electricity consumption decrease below:

- Air side equipment electricity consumption has decreased by 10.8% in November 2024 compared with the same month in 2023. This is due to the newly installed VSD for 12 units of exhaust air fan and air side equipment control optimization for casino, spectacle and kitchen.

Table 22 presents the monthly e-bus electricity consumption in MGM COTAI. The e-bus electricity consumption in November 2024 has decreased by 22.0% compared to the same month in 2023.

Table 22. COTAI - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2023	2024	% Change – Actual
January	51,261	137,030	+167.3%
February	53,640	142,468	+165.6%
March	66,865	151,047	+125.9%
April	72,669	162,250	+123.3%
May	135,032	163,600	+21.2%
June	140,991	170,402	+20.9%
July	148,375	176,644	+19.1%
August	141,507	145,579	+2.9%
September	140,705	116,650	-17.1%
October	139,616	121,385	-13.1%
November	135,280	105,531	-22.0%
December			
Total	**1,225,941**	**1,592,587**	**+29.9%**

Table 23 presents the comparison of electricity rates in the years 2023 and 2024. The electricity rate in November 2024 decreased by 1.6% compared to the same month in 2023.

Table 23. COTAI - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2023 – Electricity Rate (MOP/kWh)	Year 2024 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.241	1.241	-0.0%
February	1.245	1.244	-0.1%
March	1.242	1.242	+0.0%
April	1.230	1.250	+1.7%
May	1.231	1.249	+1.5%
June	1.228	1.250	+1.8%
July	1.229	1.239	+0.8%
August	1.228	1.240	+1.0%
September	1.230	1.240	+0.9%
October	1.221	1.200	-1.7%
November	1.223	1.203	-1.6%
December			
Average	**1.231**	**1.236**	**+0.4%**

Table 24 summarizes the monthly normalized electricity consumption. The actual electricity consumption in November 2024 decreased by 9.9% compared to 2019 normalized consumption in the same month.

Table 24. COTAI - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2024 Actual Electricity Consumption (kWh)	% Change – Normalized
January	9,166,278	9,054,720	-1.2%
February	9,055,928	8,597,400	-5.1%
March	9,881,039	9,700,320	-1.8%
April	11,260,485	10,820,760	-3.9%
May	11,419,384	11,032,920	-3.4%
June	12,039,233	11,777,640	-2.2%
July	12,681,609	12,270,000	-3.2%
August	12,725,315	12,230,400	-3.9%
September	12,344,624	11,393,400	-7.7%
October	12,071,494	10,790,160	-10.6%
November	10,685,441	9,630,120	-9.9%
December			
Total	**123,330,830**	**117,297,840**	**-4.9%**

3.2 NATURAL GAS CONSUMPTION

The actual natural gas consumption and cost in November 2024 increased by 6.8% and 2.8% respectively compared to the same month in 2023, respectively. To date, MGM COTAI has increased actual natural gas consumption by 21.9% and associated costs by 19.2%.

Table 25. COTAI - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual NG Consumption (m^3)			Actual NG Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2023	2024	% Change Actual	2023	2024	% Change Actual		
January	64,043	88,198	+37.7%	405,800	607,700	+49.8%	-37.6%	+49.1%
February	58,053	74,893	+29.0%	378,840	498,460	+31.6%	-35.8%	+48.7%
March	63,461	76,680	+20.8%	438,660	510,350	+16.3%	-41.6%	+46.0%
April	62,658	77,305	+23.4%	433,120	514,510	+18.8%	+10.3%	+28.4%
May	66,163	84,651	+27.9%	457,350	563,400	+23.2%	+8.9%	+36.8%
June	65,456	82,348	+25.8%	452,450	548,080	+21.1%	-7.1%	+37.3%
July	69,467	89,503	+28.8%	480,190	595,700	+24.1%	-1.5%	+31.5%
August	69,963	85,168	+21.7%	483,610	566,840	+17.2%	-3.0%	+34.1%
September	69,610	82,476	+18.5%	481,170	548,930	+14.1%	-7.7%	+27.9%
October	81,130	87,642	+8.0%	560,800	583,310	+4.0%	+14.9%	+18.9%
November	79,032	84,397	+6.8%	546,300	561,710	+2.8%	-1.7%	+62.4%
December								
Total	**749,036**	**913,261**	**+21.9%**	**5,118,290**	**6,098,990**	**+19.2%**	**-6.5%**	**+37.2%**

Important note: *Minimum Outdoor Air Temperature is the main variable taken into consideration on natural gas consumption normalization*

NV5 summarized the key items which could result in natural gas consumption increase below:

- Kitchen NG consumption for November 2024 has increased by 7.8% and Food cover has increased by 62.4% compared to the same month in 2023.

Table 26 presents the comparison of natural gas rates in the years 2023 and 2024. The natural gas rate in November 2024 decreased by 3.7% compared to the same month in 2023.

Table 26. COTAI - Summary of Actual Monthly Natural Gas Rate

Monthly Period	Year 2023 – Natural Gas Rate (MOP/m³)	Year 2024 – Natural Gas Rate (MOP/m³)	% Change – NG Rate
January	6.34	6.89	+8.7%
February	6.53	6.66	+2.0%
March	6.91	6.66	-3.7%
April	6.91	6.66	-3.7%
May	6.91	6.66	-3.7%
June	6.91	6.66	-3.7%
July	6.91	6.66	-3.7%
August	6.91	6.66	-3.7%
September	6.91	6.66	-3.7%
October	6.91	6.66	-3.7%
November	6.91	6.66	-3.7%
December			
Average	**6.82**	**6.68**	**-2.2%**

Table 27 summarizes the monthly normalized natural gas consumption. The actual natural gas consumption in November 2024 decreased by 28.2% compared to 2019 normalized consumption in the same month.

Table 27. COTAI - Summary of Normalized Monthly Natural Gas Consumption

Monthly Period	2019 Normalized Natural Gas Consumption (m³)	2024 Actual Natural Gas Consumption (m³)	% Change – Normalized
January	134,135	88,198	-34.2%
February	129,115	74,893	-42.0%
March	128,801	76,680	-40.5%
April	110,291	77,305	-29.9%
May	109,506	84,651	-22.7%
June	106,996	82,348	-23.0%
July	103,231	89,503	-13.3%
August	103,231	85,168	-17.5%
September	108,251	82,476	-23.8%
October	107,624	87,642	-18.6%
November	117,507	84,397	-28.2%
December			
Total	**1,258,689**	**913,261**	**-27.4%**

3.3 ISO50001 ENERGY CONSUMPTION RECORD

COTAI ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to November 2024 is **6.9%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

Table 28. COTAI ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Normalized Savings	
COTAI	2019	2025 Target	**-13%**
		Savings as of November 2024	**-6.9%**

Table 29. COTAI ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2024 Actual Energy Consumption (GJ)	% Change – Normalized
January	37,653	35,657	-5.3%
February	37,082	33,549	-9.5%
March	40,041	37,582	-6.1%
April	44,365	41,637	-6.1%
May	44,910	42,656	-5.0%
June	47,054	45,257	-3.8%
July	49,236	47,278	-4.0%
August	49,393	46,985	-4.9%
September	48,197	43,878	-9.0%
October	47,192	41,886	-11.2%
November	42,545	37,597	-11.6%
December			
Total	**487,667**	**453,962**	-6.9%

***Important note**: Factors used for unit conversion: Electricity (0.0036 GJ/kWh) and Natural Gas (0.0347 GJ/m³)*

3.4 WATER CONSUMPTION

According to Table 30, the actual water consumption and associated costs in November 2024 has decreased by 3.0% and 3.0% respectively compared to the same month in 2023 respectively. To date, MGM COTAI has increased both the actual water consumption and associated cost by 3.5% and 3.5% respectively.

Table 30. COTAI - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2023	2024	% Change Actual	2023	2024	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	54,018	64,264	+19.0%	419,790	499,180	+18.9%	+173.7%	+49.1%
February	61,317	66,870	+9.1%	476,420	519,440	+9.0%	+17.2%	+48.7%
March	61,936	65,464	+5.7%	481,140	508,610	+5.7%	+5.6%	+46.0%
April	70,617	79,200	+12.2%	548,440	615,060	+12.1%	+50.0%	+28.4%
May	80,211	76,182	-5.0%	622,920	591,560	-5.0%	-3.5%	+36.8%
June	84,354	74,796	-11.3%	654,900	580,830	-11.3%	-6.1%	+37.3%
July	94,905	96,947	+2.2%	736,800	752,580	+2.1%	-1.4%	+31.5%
August	85,434	92,595	+8.4%	663,190	718,810	+8.4%	+2.5%	+34.1%
September	83,149	83,835	+0.8%	645,490	650,830	+0.8%	+3.7%	+27.9%
October	79,151	86,778	+9.6%	614,630	673,760	+9.6%	+8.6%	+18.9%
November	77,515	75,211	-3.0%	602,010	584,090	-3.0%	-8.4%	+62.4%
December								
Total	**832,607**	**862,142**	**+3.5%**	**6,465,730**	**6,694,750**	**+3.5%**	**+5.3%**	**+37.2%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

NV5 summarized the key items which could result in water consumption decrease below:

- Irrigation water consumption for November 2024 has decreased by 27.5% compared with the same month in 2023.
- Hotel Towers water consumption for November 2024 has decreased by 3.9% compared with the same month in 2023.

Table 31 presents the comparison of water rates in the years 2023 and 2024. The water rate in November 2024 remains unchanged when compared to the same month in 2023. The average utility rate of water has decreased by 0.01% compared to 2023.

Table 31. COTAI - Summary of Actual Monthly Water Rates

Monthly Period	Year 2023 – Water Rate (MOP/m³)	Year 2024 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.771	7.768	-0.05%
February	7.770	7.768	-0.02%
March	7.768	7.769	+0.01%
April	7.766	7.766	-
May	7.766	7.765	-0.01%
June	7.764	7.766	+0.02%
July	7.764	7.763	-0.01%
August	7.763	7.763	-
September	7.763	7.763	-
October	7.765	7.764	-0.01%
November	7.766	7.766	-
December			
Average	**7.766**	**7.766**	**-0.01%**

Table 32 summarized the monthly normalized water consumption. The actual water consumption in November 2024 has decreased by 21.6% compared to the same month in 2019. To date, the average utility rate of water has decreased by 9.7% compared to 2019.

Table 32. COTAI - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2024 Actual Water Consumption (m³)	% Change – Normalized
January	72,673	64,264	-11.6%
February	68,537	66,870	-2.4%
March	74,775	65,464	-12.5%
April	82,346	79,200	-3.8%
May	86,067	76,182	-11.5%
June	89,230	74,796	-16.2%
July	96,008	96,947	+1.0%
August	97,289	92,595	-4.8%
September	91,145	83,835	-8.0%
October	100,625	86,778	-13.8%
November	95,986	75,211	-21.6%
December			
Total	**954,681**	**862,142**	**-9.7%**

3.5 COTAI YEAR 2024 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2024 and reduce the site energy and water consumption which supports MGM energy and water savings target.

Table 33. COTAI Year 2024 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset…)	On-going
2	PICCV Implementation 2024 (14 units)	On-going
3	AHU/PAU EC Fan Replacement (Phase 1 : 5 units)	On-going
4	Additional VSD for Exhaust Air Fan (Phase 2 : 12 units)	Completed
5	KEF Steps Speed Control Optimization 2024 (Chun, IRD, Hu Dao, Coast, Grill 58, Mansion)	On-going
6	Back of House Offices Lighting Replacement	Completed
7	CRO Booster Pump Replacement (Phase 2 : 15 units)	Completed
Water		
8	Extension of B2 Carpark CDP Water Recycling (Phase 11)	Completed
9	Extension of B2 Carpark CDP Water Recycling (Phase 12)	Completed

APPENDIX A

Regression Analysis

The 2019 normalized consumption modelling has remained used for tracking the agreed normalized energy savings target for MGM MACAU and MGM COTAI. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2024.

MGM MACAU

Utilities	Variables	
Electricity (kWh)	CDD (Base T=17°C) & Food Cover	
LPG (kg) / Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM COTAI

Utilities	Variables	
Electricity (kWh)	CDD (Base T=17°C), Food Cover, Room Night, E-Bus Mileage	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

APPENDIX B

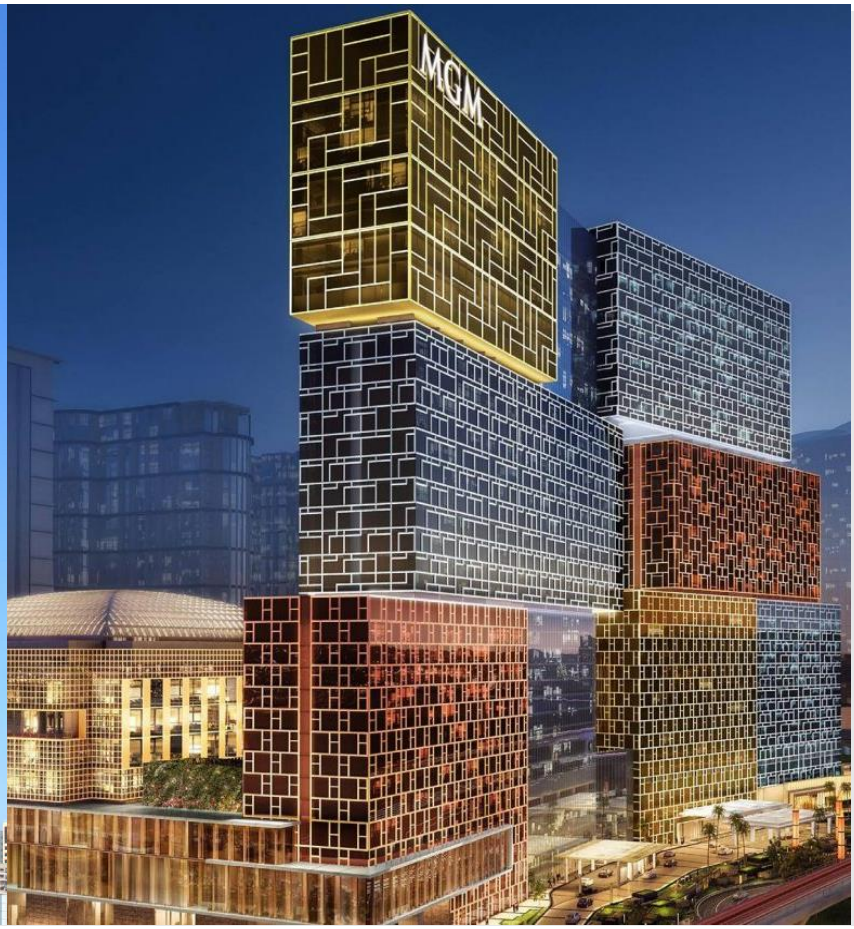
MACAU Gas Consumption Raw Data

Table 34. MACAU Gas Consumption Raw Data

Monthly Period	2023 LPG Consumption (kg)	2024 LPG Consumption (kg)	2024 NG Consumption (m^3)
January	123,108	50,003	100,712
February	78,483	-	158,122
March	77,597	-	133,119
April	77,139	-	117,800
May	80,383	-	108,947
June	71,232	-	97,924
July	67,441	-	99,596
August	71,984	-	101,187
September	69,953	-	101,015
October	78,004	-	107,043
November	77,482	-	108,030
December	117,003	-	
Total	**989,808**	**50,003**	**1,233,495**

MGM MACAU & COTAI
Resource Efficiency Management Report
860723-0004415.00
860723-0004367.00





NV5 NET ZERO

Generating Great Returns. Improving the Environment.